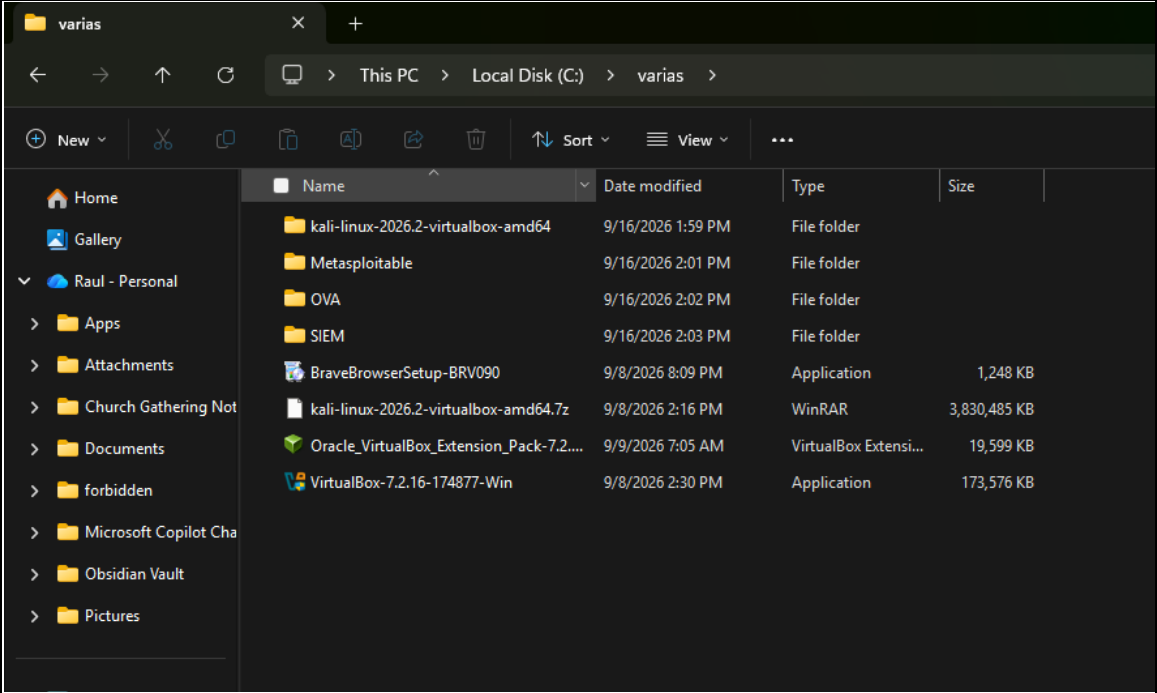
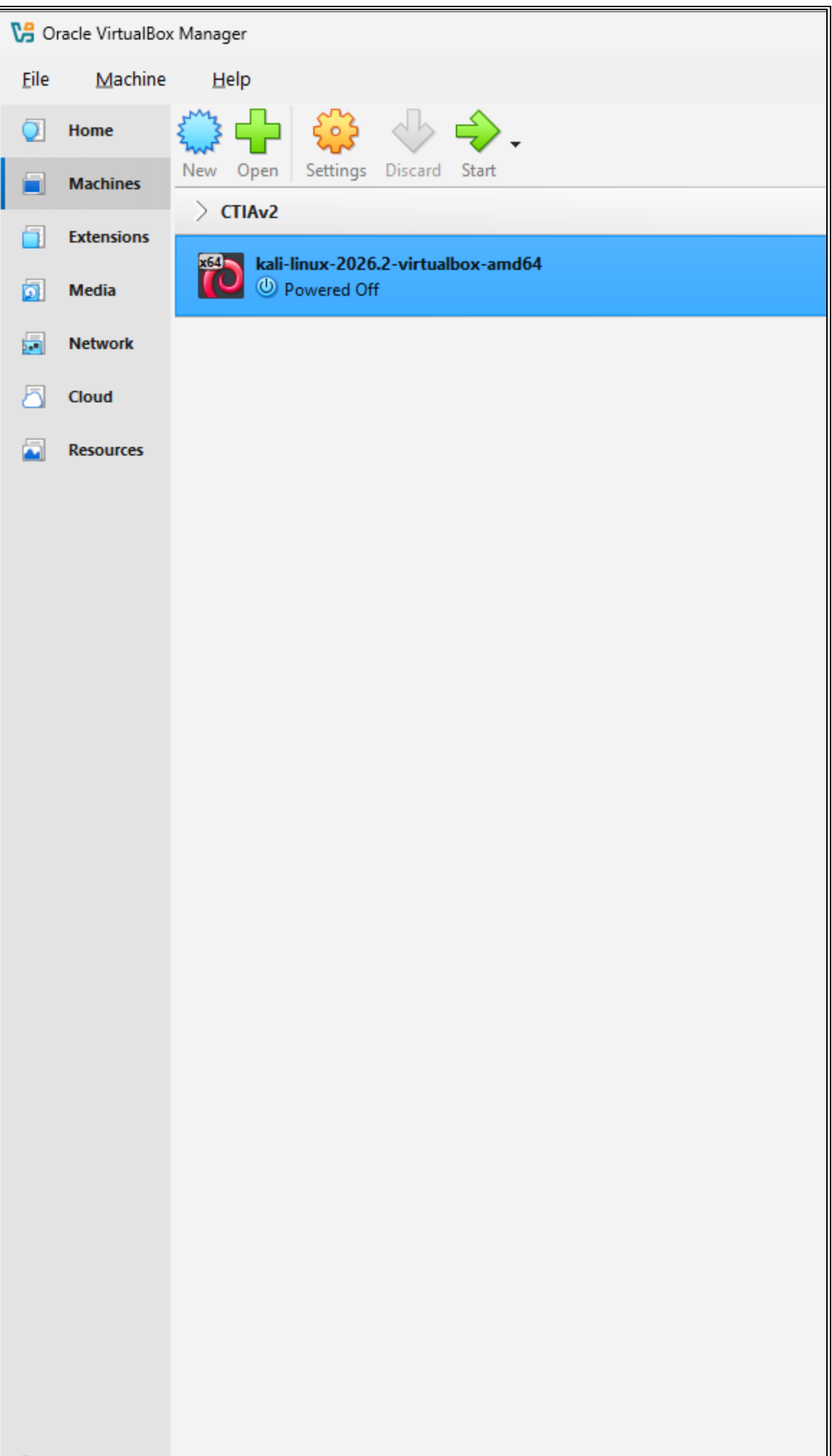


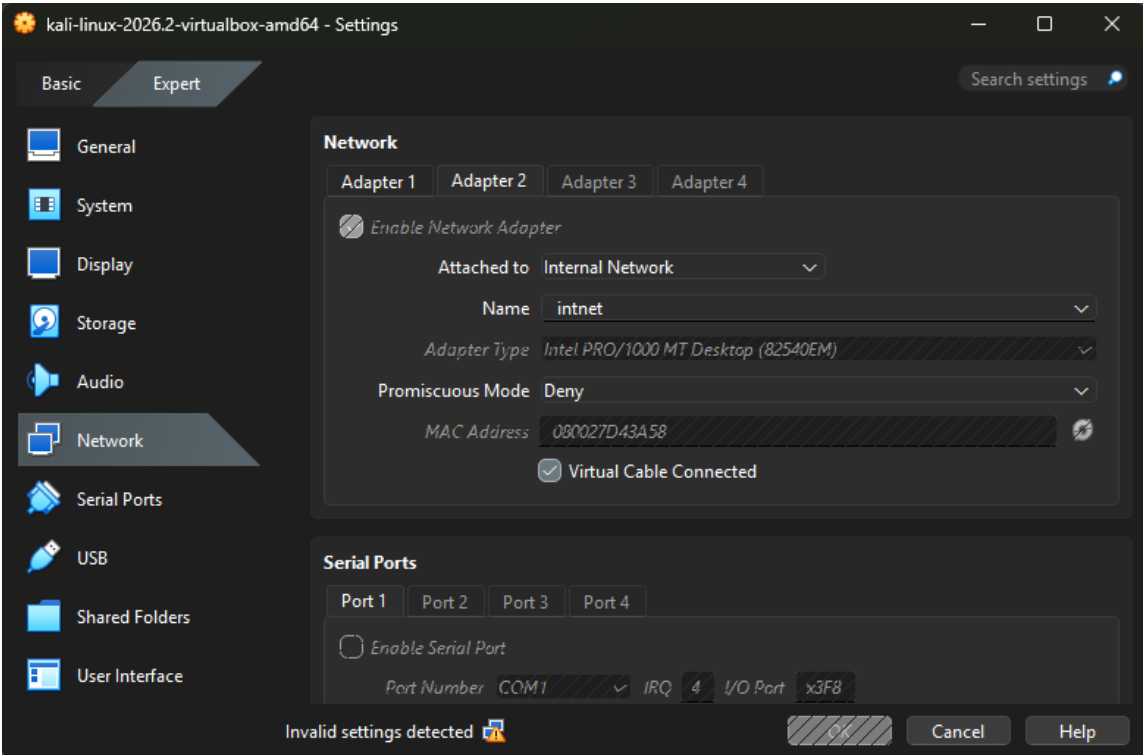
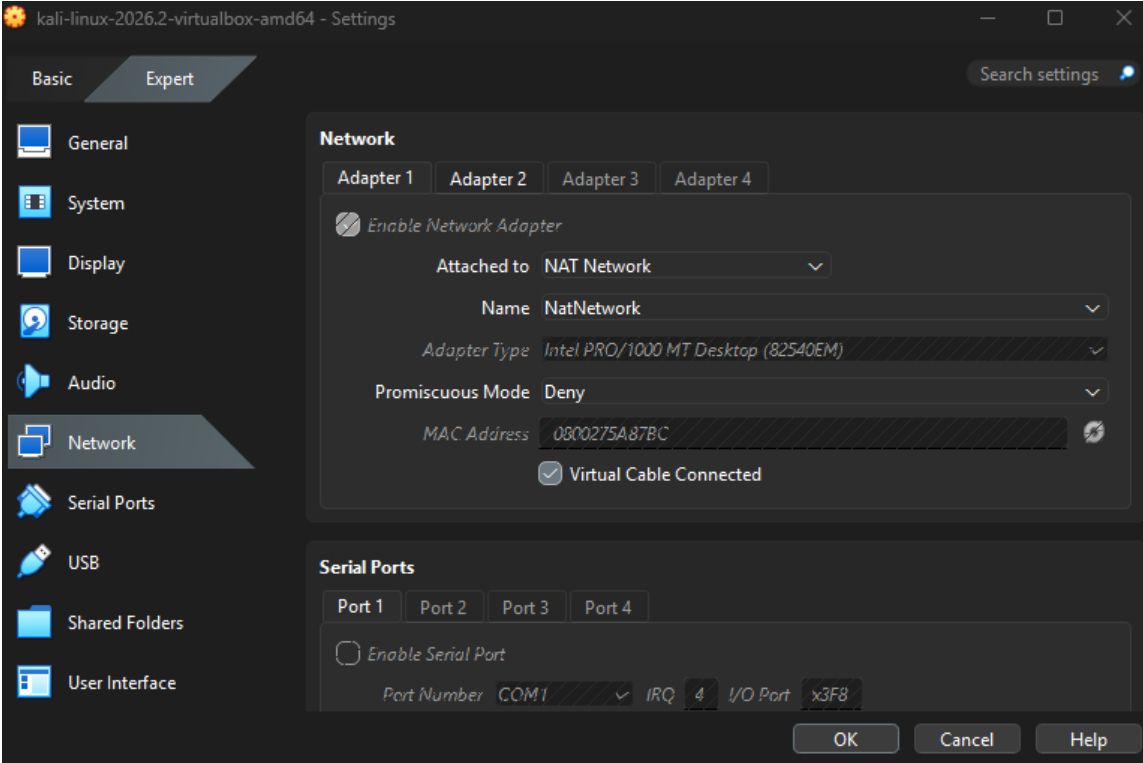
Required Evidence for Each Step

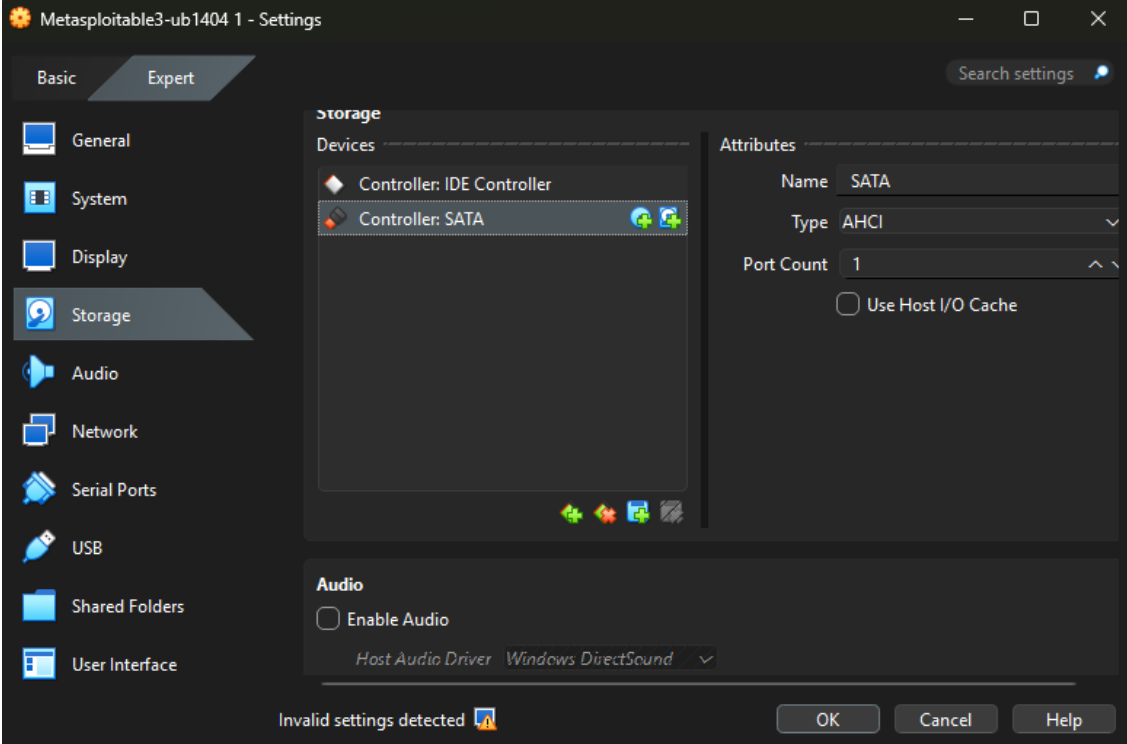
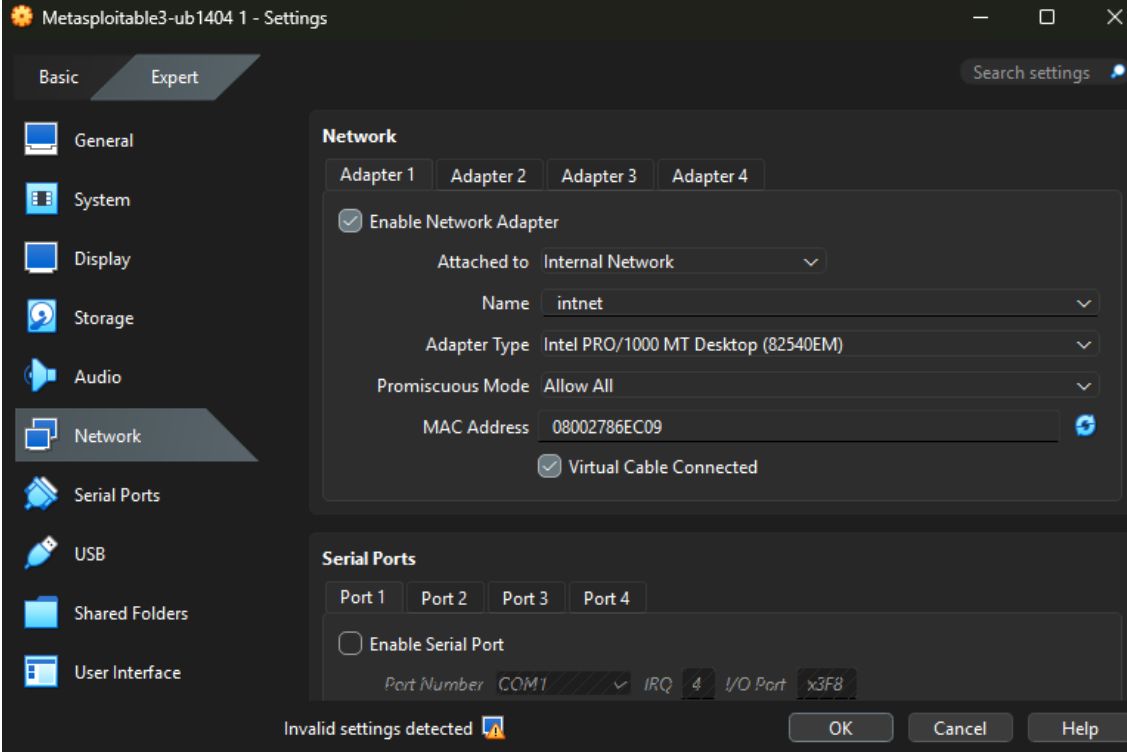
Step	What to Show	How to Capture It
Step 1	Screenshot of File Explorer showing D:\YourLastName folder with all VM folders inside (Kali, Metasploitable, SecurityOnion)	
Step 2	Not required – no permanent change to capture	–

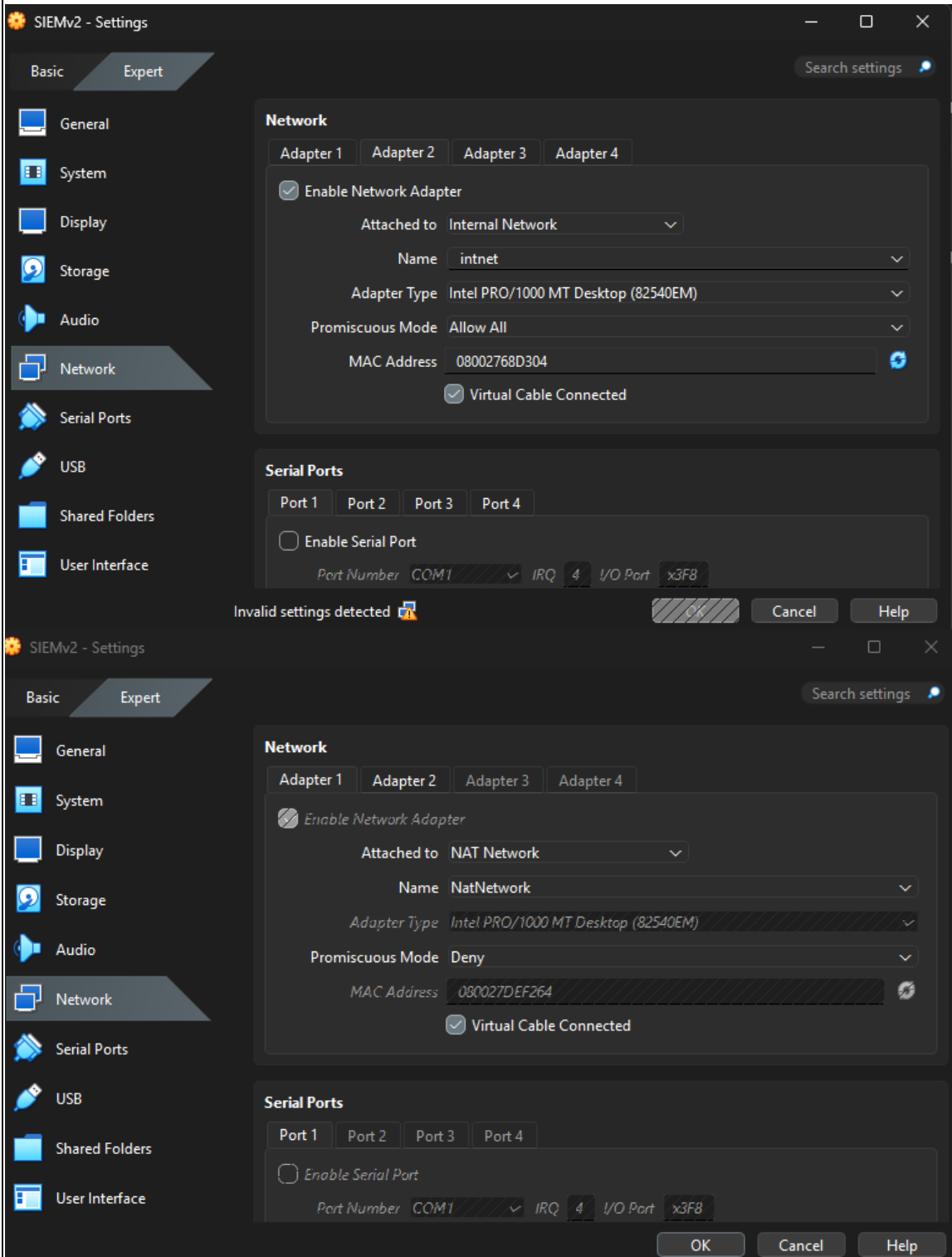
Step
3

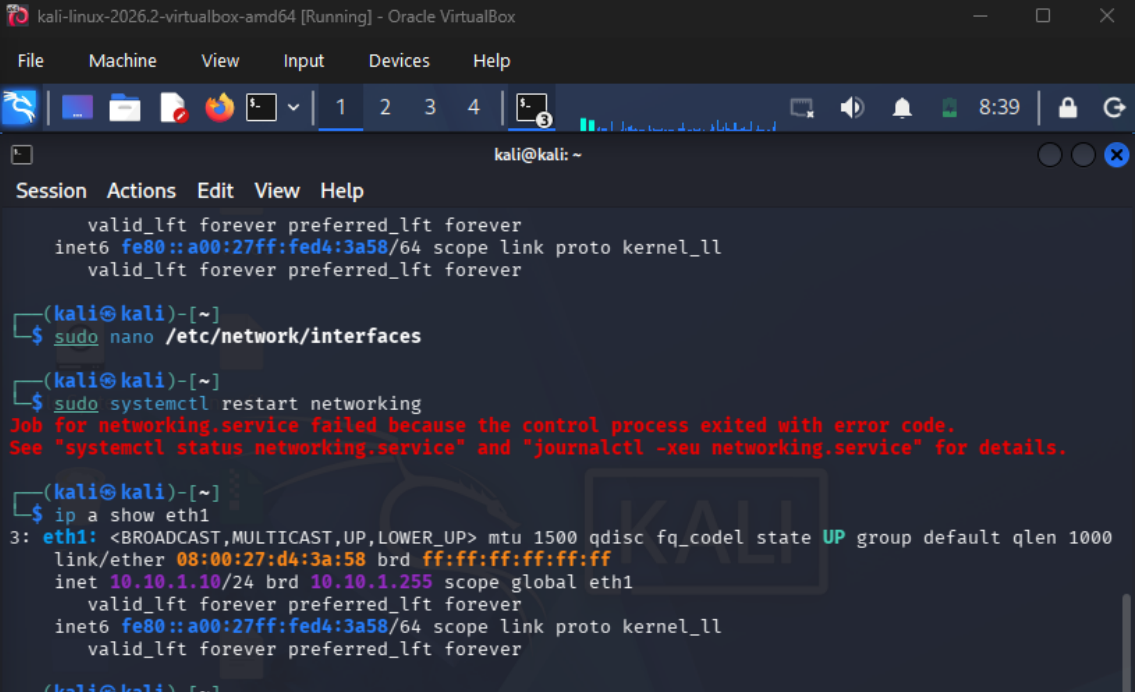
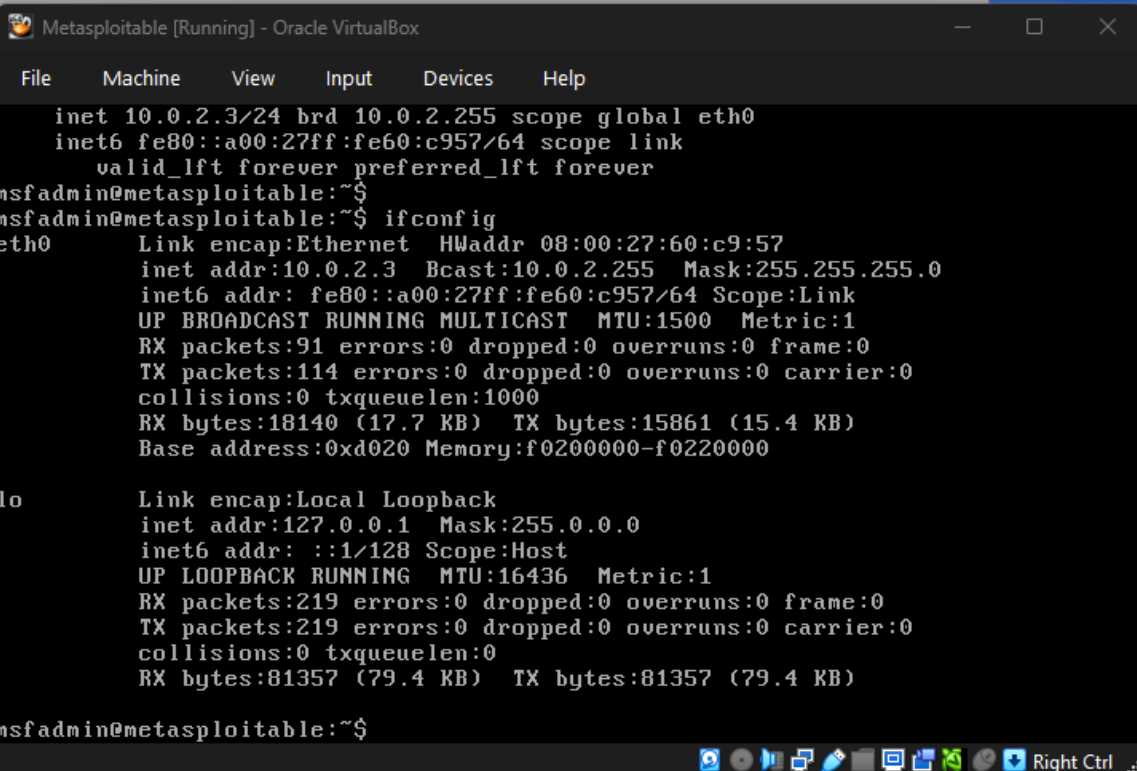
Screenshot
of **VirtualBox
Manager** sho
wing **Kali
Linux** VM in
the left list

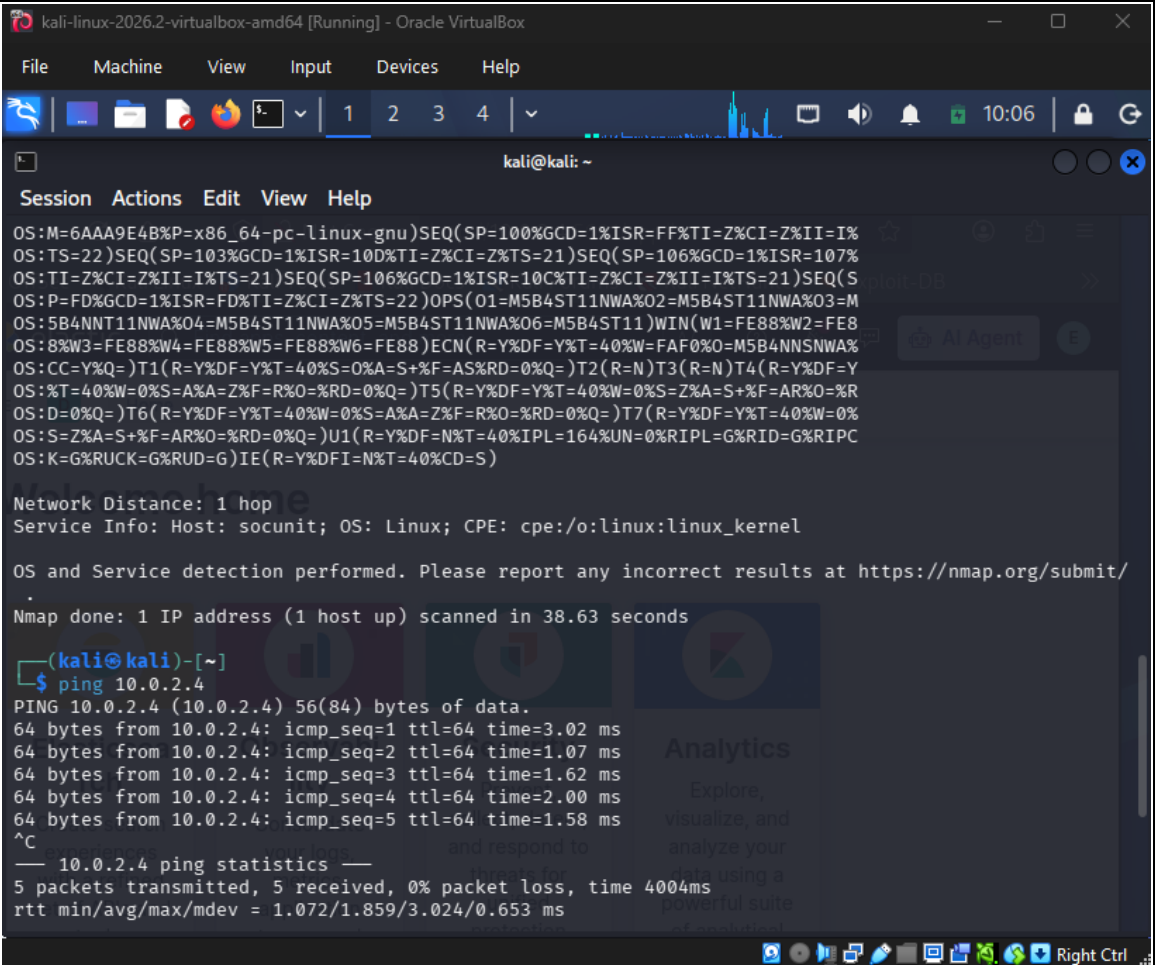


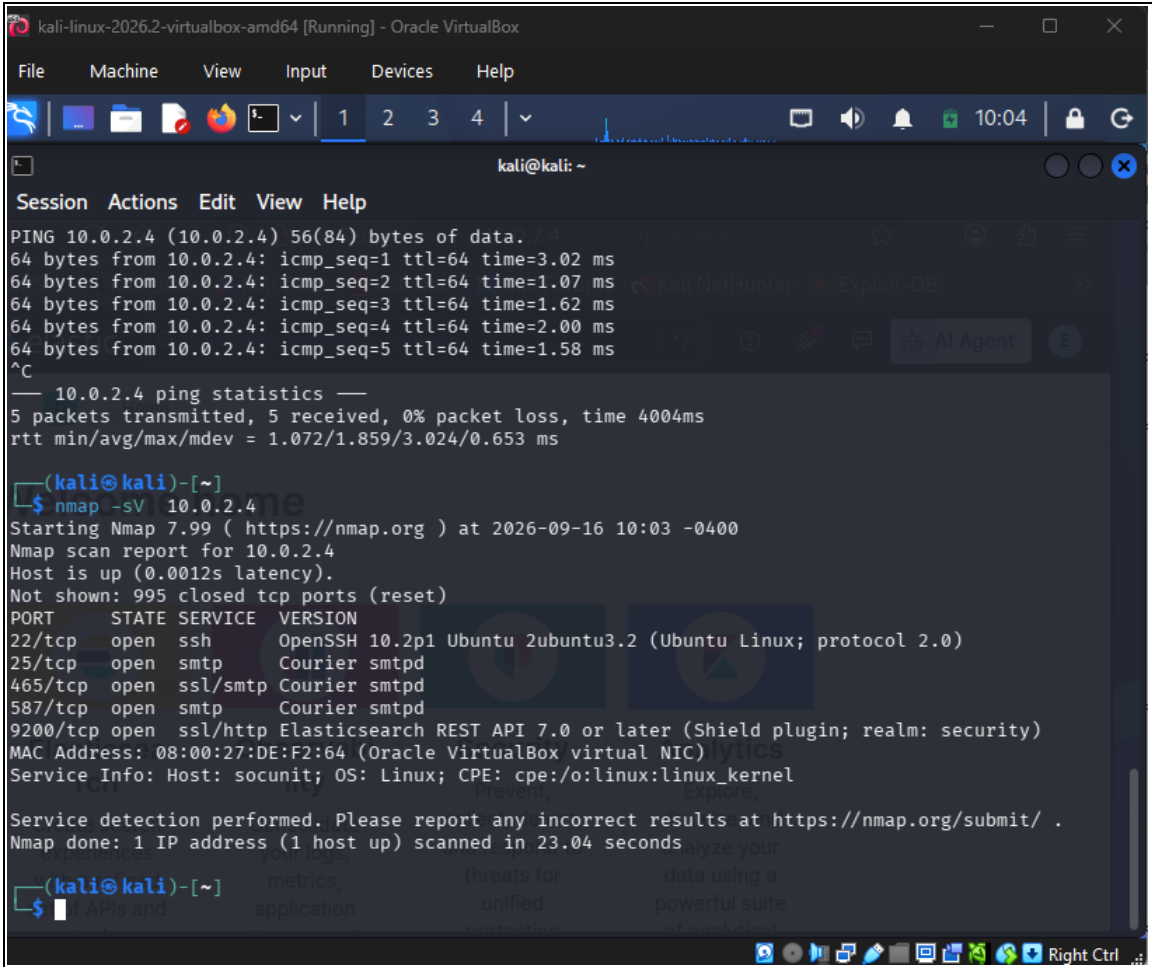
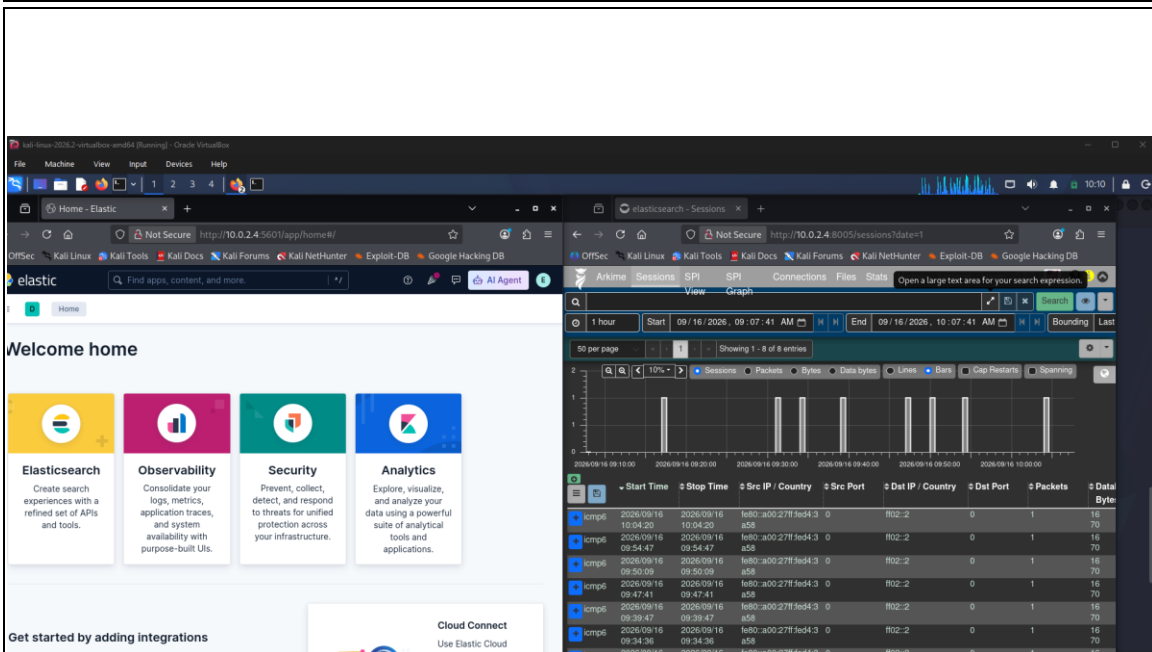
Step	What to Show	How to Capture It
Step 4	Screenshot of Kali Settings → Network tab showing Adapter 1 = NAT, Adapter 2 = Internal Network	 This screenshot shows the 'Network' settings for 'Adapter 2' in a Kali Linux virtual machine. The 'Attached to' dropdown is set to 'Internal Network'. The 'Name' is 'intnet'. The 'Adapter Type' is 'Intel PRO/1000 MT Desktop (82540EM)'. 'Promiscuous Mode' is set to 'Deny'. The 'MAC Address' is '080027D43A58'. The 'Virtual Cable Connected' checkbox is checked. A message at the bottom states 'Invalid settings detected'.  This screenshot shows the 'Network' settings for 'Adapter 1' in a Kali Linux virtual machine. The 'Attached to' dropdown is set to 'NAT Network'. The 'Name' is 'NatNetwork'. The 'Adapter Type' is 'Intel PRO/1000 MT Desktop (82540EM)'. 'Promiscuous Mode' is set to 'Deny'. The 'MAC Address' is '0800275A87BC'. The 'Virtual Cable Connected' checkbox is checked. The 'Invalid settings detected' message is no longer present.

Step	What to Show	How to Capture It
Step 5	Screenshot of Metasploitable Settings → Storage showing the .vmdk file attached and Network tab showing Adapter 1 = Internal Network	Two screenshots (Storage + Network)  

Step	What to Show	How to Capture It
Step 6	Screenshot of SIEM Settings → Network showing all 2 adapters (NAT NETWORK, Internal with promiscuous = Allow All)	 The image displays two screenshots of the SIEMv2 - Settings application, specifically the 'Network' tab. The top screenshot shows the 'Adapter 1' settings, where 'Enable Network Adapter' is checked, 'Attached to' is set to 'Internal Network', 'Name' is 'intnet', 'Adapter Type' is 'Intel PRO/1000 MT Desktop (82540EM)', 'Promiscuous Mode' is 'Allow All', and 'MAC Address' is '08002768D304'. The bottom screenshot shows the 'Adapter 2' settings, where 'Enable Network Adapter' is checked, 'Attached to' is set to 'NAT Network', 'Name' is 'NatNetwork', 'Adapter Type' is 'Intel PRO/1000 MT Desktop (82540EM)', 'Promiscuous Mode' is 'Deny', and 'MAC Address' is '080027DEF264'. Both screenshots show the 'Serial Ports' section with 'Enable Serial Port' unchecked, 'Port Number' set to 'COM1', 'IRQ' set to '4', and 'I/O Port' set to 'x3F8'. An 'Invalid settings detected' message is visible in the bottom screenshot.

Step	What to Show	How to Capture It
Step 7	Screenshot of Kali terminal after running <code>ip a show eth1</code> showing IP address <code>10.10.1.10</code>	 <pre> kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox File Machine View Input Devices Help kali@kali: ~ Session Actions Edit View Help valid_lft forever preferred_lft forever inet6 fe80::a00:27ff:fed4:3a58/64 scope link proto kernel_ll valid_lft forever preferred_lft forever (kali@kali)-[~] \$ sudo nano /etc/network/interfaces (kali@kali)-[~] \$ sudo systemctl restart networking Job for networking.service failed because the control process exited with error code. See "systemctl status networking.service" and "journalctl -xeu networking.service" for details. (kali@kali)-[~] \$ ip a show eth1 3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000 link/ether 08:00:27:d4:3a:58 brd ff:ff:ff:ff:ff:ff inet 10.10.1.10/24 brd 10.10.1.255 scope global eth1 valid_lft forever preferred_lft forever inet6 fe80::a00:27ff:fed4:3a58/64 scope link proto kernel_ll valid_lft forever preferred_lft forever (kali@kali)-[~] </pre>
Step 8	Screenshot of Metasploitable terminal after running <code>ifconfig</code> showing IP address <code>10.10.1.20</code>	 <pre> Metasploitable [Running] - Oracle VirtualBox File Machine View Input Devices Help inet 10.0.2.3/24 brd 10.0.2.255 scope global eth0 inet6 fe80::a00:27ff:fe60:c957/64 scope link valid_lft forever preferred_lft forever msfadmin@metasploitable:~\$ msfadmin@metasploitable:~\$ ifconfig eth0 Link encap:Ethernet HWaddr 08:00:27:60:c9:57 inet addr:10.0.2.3 Bcast:10.0.2.255 Mask:255.255.255.0 inet6 addr: fe80::a00:27ff:fe60:c957/64 Scope:Link UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1 RX packets:91 errors:0 dropped:0 overruns:0 frame:0 TX packets:114 errors:0 dropped:0 overruns:0 carrier:0 collisions:0 txqueuelen:1000 RX bytes:18140 (17.7 KB) TX bytes:15861 (15.4 KB) Base address:0xd020 Memory:f0200000-f0220000 lo Link encap:Local Loopback inet addr:127.0.0.1 Mask:255.0.0.0 inet6 addr: ::1/128 Scope:Host UP LOOPBACK RUNNING MTU:16436 Metric:1 RX packets:219 errors:0 dropped:0 overruns:0 frame:0 TX packets:219 errors:0 dropped:0 overruns:0 carrier:0 collisions:0 txqueuelen:0 RX bytes:81357 (79.4 KB) TX bytes:81357 (79.4 KB) msfadmin@metasploitable:~\$ </pre>

Step	What to Show	How to Capture It
Step 9	Screenshot of Kali terminal showing successful ping 10.10.1.20 output (at least 3 reply lines)	 The screenshot shows a Kali Linux terminal window with the following content: <pre> kali@kali: ~ Session Actions Edit View Help OS: M=6AAA9E4B%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=FF%TI=Z%CI=Z%II=I% OS: TS=22)SEQ(SP=103%GCD=1%ISR=10D%TI=Z%CI=Z%TS=21)SEQ(SP=106%GCD=1%ISR=107% OS: TI=Z%CI=Z%II=I%TS=21)SEQ(SP=106%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=21)SEQ(S OS: P=FD%GCD=1%ISR=FD%TI=Z%CI=Z%TS=22)OPS(O1=M5B4ST11NWA%O2=M5B4ST11NWA%O3=M OS: 5B4NNT11NWA%O4=M5B4ST11NWA%O5=M5B4ST11NWA%O6=M5B4ST11)WIN(W1=FE88%W2=FE8 OS: 8%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B4NNSNWA% OS: CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y OS: %T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%R OS: D=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0% OS: S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPC OS: K=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S) Network Distance: 1 hop Service Info: Host: socunit; OS: Linux; CPE: cpe:/o:linux:linux_kernel OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ . Nmap done: 1 IP address (1 host up) scanned in 38.63 seconds (kali@kali)-[~] \$ ping 10.0.2.4 PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data: 64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=3.02 ms 64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.07 ms 64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=1.62 ms 64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=2.00 ms 64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=1.58 ms ^C — 10.0.2.4 ping statistics — 5 packets transmitted, 5 received, 0% packet loss, time 4004ms rtt min/avg/max/mdev = 1.072/1.859/3.024/0.653 ms </pre>

Step	What to Show	How to Capture It
Step 10	Screenshot of Kali terminal showing nmap -sV 10.10.1.20 output with open ports listed	
Step 11	Two things: (a) Screenshot of SIEM web interface (Kibana) logged in (b) Screenshot showing search results for host: 10.10.1.20 (c) Screenshot of SIEM web interface (Arkime) logged in	

Written Answers

Answer these **in the same document** after your screenshots:

1. What command did you use in Kali to set the static IP on the second interface?

Create a configuration file for `eth1`:

```
sudo nano /etc/network/interfaces
```

- Add these lines at the end:
- `auto eth1`
- `iface eth1 inet static`
- `address 10.10.1.10`
`netmask 255.255.255.0`
- **Save:** `Ctrl+O`, `Enter`, `Ctrl+X`

2. What does `PROMISC` mean in the output of `ip a show eth1` on Security Onion?

PROMISC mode configures a network adapter to log all passing data packets rather than filtering out traffic addressed to other devices. Security tools like Suricata and SIEM systems rely on this setup to maintain full visibility for network monitoring and threat detection.

3. List any **three open ports** you saw on Metasploitable from the Nmap scan.

Port 22/TCP Service SSH

Port 25/TCP Service smtp

Port 9200/TCP ssl/http

4. In SIEM Kibana web interface, what **Alert Signature Name** appeared for your Nmap scan?

ET SCAN Potential Nmap Scan